

DODO: Multimodal Dataset Creation Tool

Table of Contents

1. [Overview](#)
2. [Installation](#)
3. [Quick Start](#)
4. [Commands](#)
5. [Security Features](#)
6. [Data Formats](#)
7. [Examples](#)
8. [API Reference](#)
9. [Troubleshooting](#)

Overview

DODO is a powerful command-line tool that transforms raw robotics logs into structured, ML-ready multimodal datasets. It automatically aligns sensor data (camera, IMU, lidar, actions) into synchronized frames and generates comprehensive metadata for computer vision, robotics, and multimodal AI training.

Key Features

- **Multimodal Data Processing:** Handle camera, IMU, lidar, and action data
- **Automatic Alignment:** Synchronize sensor data across different frequencies
- **Duplicate Removal:** Filter out redundant data with advanced algorithms
- **Security:** Enterprise-grade authentication and encryption
- **Multiple Formats:** Support for rosbag, JSON, JSONL, and mock logs
- **Validation:** Built-in dataset integrity checking

Installation

Prerequisites

- Python 3.8 or higher
- Required dependencies: typer, click, pydantic, cryptography, numpy

Install from Source

```
bash git clone <repository-url> cd dodo pip install -e .
```

Dependencies

```
bash pip install typer click pydantic cryptography numpy
```

Quick Start

1. Create User Account

```
bash dodo register --username myuser --email user@example.com
```

2. Login

```
bash dodo login --username myuser
```

3. Create Project

```
bash dodo init my_dataset --task navigation
```

4. Import Data

```
bash dodo import robot_data.bag --project my_dataset
```

5. Generate Metadata

```
bash dodo generate-metadata --project my_dataset
```

6. Filter Duplicates

```
bash dodo filter frames --project my_dataset dodo filter messages --project my_dataset
```

7. Export Dataset

```
bash dodo export output/ --project my_dataset
```

Commands

Project Management

init

Create a new DODO project. `bash dodo init <project_name> [--task TASK] [--version VERSION]` - `project_name`: Name of the DODO dataset project - `--task, -t`: Dataset task label (default: navigation) - `--version`: Project version (default: 0.1.0)

list

List all DODO projects. `bash dodo list [--path PATH]` - `--path, -p`: Path to search for DODO projects (default: current directory)

delete

Delete a DODO project. `bash dodo delete <project_name> [--force] - project_name` :
Name of the DODO project to delete - `--force, -f` : Force deletion without confirmation

Data Operations

import

Import robot logs into a DODO project. `bash dodo import <file_path> [--project PROJECT] - file_path` : Path to rosbag, JSON, JSONL, or mock log - `--project, -p` : DODO project root

generate-metadata

Generate aligned frame metadata. `bash dodo generate-metadata [--project PROJECT] - --project, -p` : DODO project root

filter

Remove duplicate data from dataset (requires authentication). `bash dodo filter <filter_type> [--project PROJECT] - filter_type` : 'frames' or 'messages' - `--project, -p` : DODO project root

export

Export dataset to structured format. `bash dodo export <output_path> [--project PROJECT] [--overwrite] - output_path` : Destination path for the structured dataset - `--project, -p` : DODO project root - `--overwrite, -f` : Overwrite existing output directory

Data Inspection

show-metadata

Display DODO dataset metadata. `bash dodo show-metadata [--project PROJECT] - --project, -p` : DODO project root

validate

Validate dataset integrity. `bash dodo validate [--project PROJECT] - --project, -p` : DODO project root

Security Commands

register

Create a new user account. `bash dodo register --username USERNAME [--email EMAIL] - --username, -u` : Username for new account - `--email, -e` : Email address (optional)

login

Login to DODO with authentication. `bash dodo login --username USERNAME [--password`

PASSWORD] - --username, -u : Username for login - --password, -p : Password (will prompt if not provided)

Logout

Logout from current DODO session. `bash dodo logout`

whoami

Show current logged in user. `bash dodo whoami`

change-password

Change user password. `bash dodo change-password [--username USERNAME] - --username, -u : Username (defaults to current user)`

Security Features

Authentication System

- **User Registration:** Secure account creation with password validation
- **Session Management:** Time-limited sessions (1 hour expiration)
- **API Keys:** Unique secure API keys for each user
- **Password Security:** PBKDF2 key derivation with 100,000 iterations

Encryption

- **Data Encryption:** AES-256 encryption for sensitive data
- **Secure Storage:** Encrypted credential storage
- **HMAC Protection:** Timing attack prevention
- **Salt Management:** Unique salts for password hashing

Access Control

- **Command Protection:** Sensitive operations require authentication
- **Session Validation:** Automatic session expiration
- **API Key Verification:** Secure API access control

Data Formats

Input Formats

- **ROSBAG:** Robot Operating System bag files
- **JSON:** Structured JSON logs
- **JSONL:** JSON Lines format
- **Mock Logs:** Synthetic test data

Output Formats

- **Aligned Frames:** Synchronized multimodal data
- **Metadata:** Comprehensive dataset information
- **Structured Export:** ML-ready format

Sensor Types

- **Camera:** Image data (compressed/raw)
- **IMU:** Inertial measurement unit data
- **Lidar:** Point cloud data
- **Action:** Robot command/control data
- **Odometry:** Position and orientation data

Examples

Basic Workflow

```
```bash
```

## 1. Register and login

---

```
dodo register --username researcher --email lab@university.edu dodo login --username researcher
```

## 2. Create project

---

```
dodo init robot_navigation --task navigation
```

## 3. Import real data

---

```
dodo import turtlebot_data.bag --project robot_navigation
```

## 4. Process and clean data

---

```
dodo generate-metadata --project robot_navigation dodo filter frames --project robot_navigation dodo filter messages --project robot_navigation
```

## 5. Validate and export

---

```
dodo validate --project robot_navigation dodo export clean_dataset/ --project robot_navigation --overwrite ```
```

### Advanced Filtering

---

```
```bash
```

Remove duplicate frames (keeps unique synchronized data)

```
dodo filter frames --project my_dataset
```

Remove duplicate messages (cleans raw sensor data)

```
dodo filter messages --project my_dataset ```
```

Data Inspection

```
```bash
```

## View dataset overview

---

```
dodo show-metadata --project my_dataset
```

## Check data integrity

---

```
dodo validate --project my_dataset ```
```

### API Reference

---

#### Security Manager

---

The security system provides the following APIs:

##### Authentication

```
```python from dodo.security import security_manager
```

Create user

```
security_manager.create_user(username, password, email)
```

Authenticate

```
user_data = security_manager.authenticate(username, password)
```

Get current user

```
current_user = security_manager.get_current_user()
```

Logout

```
security_manager.logout() ````
```

Encryption

```
````python
```

## Encrypt data

---

```
encrypted = security_manager.encrypt_data(data, password)
```

## Decrypt data

---

```
decrypted = security_manager.decrypt_data(encrypted, password) ````
```

### API Key Management

```
````python
```

Get API key

```
api_key = security_manager.get_user_api_key(username)
```

Verify API key

```
username = security_manager.verify_api_key(api_key) ````
```

Data Models

```
````python from dodo.models import AlignedFrame, ImportedMessage, Metadata
```

## Aligned frame structure

---

```
frame = AlignedFrame(timestamp=1234567890.0, camera="path/to/image.jpg", imu={
 "linear_acceleration": [0.1, 0.0, 9.8]}, action=[0.2, 0.0])
```

## Message structure

---

---

```
message = ImportedMessage(timestamp=1234567890.0, topic="/imu", modality="imu",
payload={"data": "sensor_data"}) ````
```

## Troubleshooting

---

### Common Issues

---

#### Authentication Errors

**Problem:** "Authentication required" error **Solution:** Run `dodo login` with valid credentials

#### Session Expired

**Problem:** Commands fail after inactivity **Solution:** Run `dodo login` again to refresh session

#### Import Failures

**Problem:** Cannot import rosbag files **Solution:** Ensure ROS dependencies are installed and file is valid

#### Memory Issues

**Problem:** Large datasets cause memory errors **Solution:** Process data in chunks or increase system memory

#### Permission Errors

**Problem:** Cannot write to project directories **Solution:** Check file permissions and run with appropriate user

### Debug Mode

---

Enable verbose logging: `bash export DODO_DEBUG=1 dodo <command>`

### Configuration Files

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- User data: `~/ .dodo/`
- Users database: `~/ .dodo/users.json`
- Session data: `~/ .dodo/session.json`
- Project config: `<project>/dodo.json`

### Getting Help

---

```
````bash
```

Show all commands

dodo --help

Show command help

dodo --help

Show current user

dodo whoami ``

Best Practices

Security

1. Use strong passwords (8+ characters, mixed case, numbers, symbols)
2. Never share API keys
3. Logout when finished working
4. Change passwords regularly
5. Use different passwords for different services

Data Management

1. Validate datasets before processing
2. Filter duplicates to reduce noise
3. Export with descriptive names
4. Keep original raw data as backup
5. Document data sources and processing steps

Performance

1. Process large datasets in batches
2. Use SSD storage for faster I/O
3. Monitor system resources during processing
4. Clean temporary files regularly

Version History

Current Version: 1.0.0

- Initial release with core functionality
- Security features implemented
- Multi-format data support
- Advanced filtering capabilities

Planned Features

- Web-based interface
- Cloud storage integration
- Advanced data visualization
- Machine learning pipeline integration
- Real-time data streaming support

Support

For issues, questions, or contributions: - Documentation: Available in this guide - Commands: Use `dodo --help` for quick reference - Security: Contact system administrator for account issues

This documentation covers all DODO features as of version 1.0.0. For the latest updates, check the official repository.